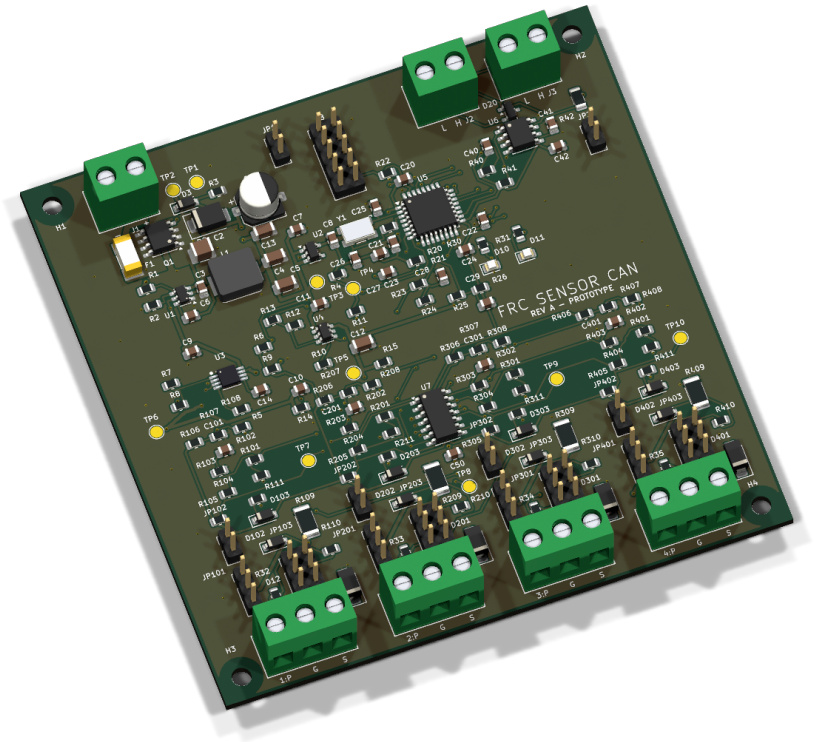
FRC
Sensor CAN

Revision A · Prototype engineering and assembly guide

Four industrial / logic sensor inputs
Protected battery and regulated 5 V sensor power



2026-09-07

Editable KiCad files, DigiKey BOM, firmware, robot examples and verification records accompany this guide. No physical prototype has been tested. Electrical ratings are design targets pending the included qualification plan.

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1 Product overview

Source: README.md

An original four-channel interface for industrial photoelectric sensors and 3.3 V / 5 V logic, with a native roboRIO CAN connection. The same nominal 12 V robot supply powers the board and its sensors. Each sensor port selects protected battery voltage or regulated 5 V; both rails can operate together.

This repository contains a routed, editable KiCad design, complete DigiKey BOM, firmware, Java/C++ robot interfaces, manufacturing files, and engineering checks. **The design has not been physically built or bench qualified.** Treat the electrical numbers below as prototype design targets, then perform the supplied bring-up tests before robot use.

Item	Revision A
Board input	6–18 V DC target, from one protected robot power-distribution branch
Sensor inputs	Four, 0–30 V DC positive signals; selectable industrial or logic thresholds
Sensor types	Three-wire DC NPN, PNP, push-pull; 3.3 V open-drain; dry contacts
Logic mode	LOW ≤0.8 V, HIGH ≥2.4 V; supports 3.3 V and 5 V logic
Industrial mode	LOW ≤4 V, HIGH ≥7 V; suitable for compatible 10–30 V sensor outputs
Sensor power	300 mA usable target **per rail, shared across all four ports**; simultaneous 5 V and battery rails
Communication	1 Mbit/s Classical CAN, documented 29-bit Team Use protocol
Data	Debounced digital states, rising-edge counts, supply diagnostics, rail faults
Construction	96 × 96 mm, two layers, 1.6 mm FR-4, 1 oz copper; top SMD and through-hole terminals
Assembly	Mostly 0805 or larger passives; LQFP32, SOIC and SOT packages; one exposed-pad IC needs heatbed/reflow
Parts estimate	**US$49.61** at checked quantity-one prices, including 13 shunts; PCB, shipping, tax and tools extra

The MCU is an STM32F042K6T6 with a CAN **controller**. The separate TCAN3413DR provides the CAN **transceiver**. The leaded MCU package, simple firmware, and checked stock made this a useful choice for hand assembly. This board uses its own protocol and does not appear as a CTRE device in Phoenix or Tuner.

Start with the printable handbook or the offline sensor setup worksheet. The worksheet produces jumper settings and a provisioning command; it does not connect to the board.

Project files:

• Design plan and research method, final architecture, and electrical review.

• KiCad project, schematic PDF, and PCB.

• Interactive assembly BOM, DigiKey purchasing files, assembly guide, and programming/mounting accessories.

• Wiring and jumper settings, printable connector/jumper card, firmware instructions, and CAN protocol.

• Robot integration, bench bring-up checklist, and validation status.

• Fabrication files, mechanical STEP model, 1:1 mounting template, and board render.

Important operating limits: battery output follows the battery; it does not generate regulated 12 V or 24 V. A sensor requiring 10 V can stop working during a robot battery sag while this interface continues running. The default logic mode does not guarantee 1.8 V recognition. PWM, quadrature, IO-Link, analog measurement,

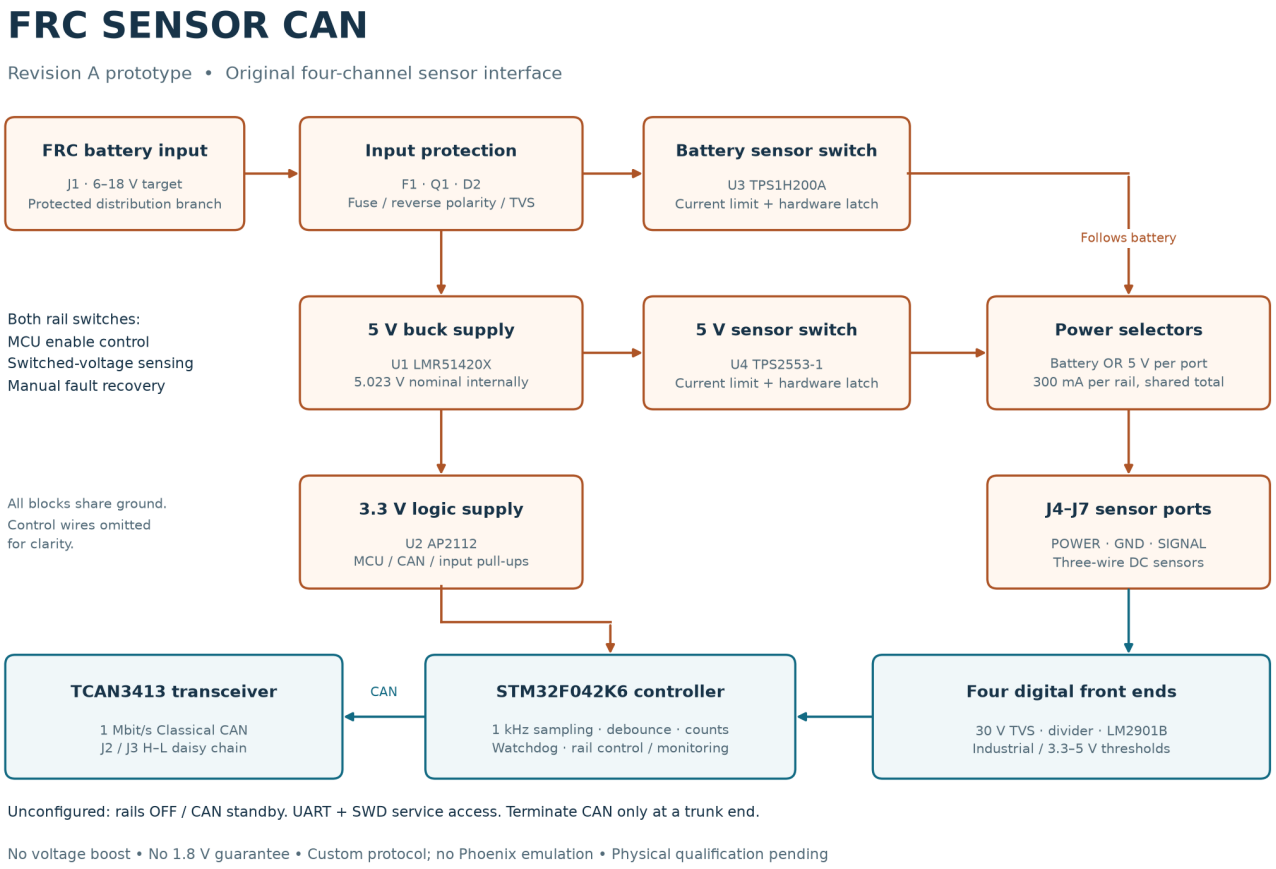
open-wire diagnosis, and CAN FD are outside this revision's implemented feature set. Keep external supply voltage off the sensor POWER pins.

All selected purchase lines had stock in the DigiKey listings checked on 2026-09-07. Those listings are dated observations, not reserved inventory or a guarantee of availability in every country. Recheck exact order codes before purchase. See sources and third-party notices.

2 Architecture and design choices

Source: docs/ARCHITECTURE.md

The design favors inspectable solder joints, explicit sensor-mode selection, and fault containment. It uses four independent digital front ends and two shared, separately switched sensor-power rails. A two-layer 96 mm square board leaves room for an iron, tweezers, test probes, screw terminals and mounting holes.



A vector PDF of this block diagram is included for printing and sharing.

The functional reference was CTRE CANdi's sensor-to-CAN role and its simultaneous battery/5 V outputs. This is an original circuit and protocol. The reference product also supports capabilities outside this revision, including PWM and quadrature decoding. CTRE CANdi

Block	Exact selected part	Reason
MCU	STM32F042K6T6	CAN controller; LQFP32 with 0.8 mm pitch; enough flash, RAM and GPIO
CAN PHY	TCAN3413DR	3.3 V, SOIC8, standby default, dominant timeout and bus-fault protection
Input protection	045101.5MRL, AO4485, MMSZ12T1G, SMBJ18A-E3/52	Fuse, low-loss reverse-polarity MOSFET, gate clamp and transient suppression
Buck regulator	LMR51420XDDCR	Stocked 500 kHz PFM synchronous regulator; 36 V capability; six exposed leads
3.3 V regulator	AP2112K-3.3TRG1	Common SOT25 regulator; logic powered ahead of sensor switches
5 V sensor switch	TPS2553DBVR-1	Adjustable current limit and hardware latch-off

Block	Exact selected part	Reason
Battery sensor switch	TPS1H200AQDGNRQ1	Adjustable current limit, configurable hardware latch delay, digital fault output
Input comparator	LM2901BIDR	Four ground-sensing comparators in SOIC14, powered from protected battery
Signal protection	SMAJ30A, BAT46W-E3-18	Positive signal transient clamp; 100 V reverse-blocking bias diodes
Terminals	Wurth 691137710002 / 691137710003	Exact 5.00 mm pitch; screw connection; through-hole assembly

An ESP32-C3 would still require an external CAN transceiver: its TWAI peripheral is the protocol controller. The F042 avoids unused radio functions and is easier to inspect than a bare QFN. An ESP32 module is a possible different implementation, but changing the MCU would also require new layout, firmware and verification. Espressif TWAI documentation

Power path

J1 feeds the onboard fuse, then Q1 with its drain toward the input and source toward the protected rail. This orientation passes correctly polarized power through the body diode during startup; negative input polarity leaves the MOSFET off. D3 limits gate-source voltage and R3 pulls the gate toward ground. An enhanced MOSFET conducts in both directions, so this circuit is not a reverse-current blocker from externally powered outputs.

D2, C1, C2 and C3 suppress and decouple the protected input. U1/L1/C4/C5/C13 produce approximately 5.023 V internally. R1 is **101 kΩ, 0.1%, 25 ppm/°C** and R2 is **13.7 kΩ, 0.1%, 25 ppm/°C**. Three 22 µF output capacitors account for substantial DC-bias capacitance loss. U2 supplies the 3.3 V logic rail.

U3 supplies SENSOR_VBAT and U4 supplies SENSOR_5V. Their current limits sit above the intended 300 mA continuous load per rail. The limits are shared, so a short on one sensor disables the other sensors using that same rail. Neither rail is for motors, solenoids or other actuators. Large sensor startup capacitance can trip a rail even when steady current is acceptable.

C14 is 33 nF, setting approximately 11 ms nominal battery hardware latch delay. R23 senses the switched battery output, and the second ADC channel senses switched 5 V. Both update at 1 kHz. An enabled rail below a measured 4.5 V latches off after 5 ms startup blanking; ADC failure latches both. U3 also reports healthy loads below 10–20 mA as open load, so its digital FAULT alone is deliberately not used to disconnect battery sensors. Autonomous hardware current and thermal protection remain active. R4 is a 1 kΩ logic bleed and R14 a 10 kΩ battery-output bleed to control off-state leakage.

Hardware pulldowns keep both sensor enables low during reset. Factory firmware without valid configuration keeps sensor rails off and the CAN transceiver in standby. Configuration is stored only by an explicit service command. Overloads latch off; recovery requires an explicit command and at least 100 ms off time. A watchdog reset starts with both rails latched off. Low logic voltage, clock failure and a stalled foreground force a safe shutdown/reset.

Input path

Each SIGNAL terminal has a 30 V standoff TVS, two 110 kΩ series resistors, a 20 kΩ divider resistor, 910 kΩ positive feedback, a 1 nF C0G filter and a comparator output pulled to 3.3 V through 10 kΩ. The comparator reference uses 100 kΩ / 17.8 kΩ; closing the threshold jumper adds 7.15 kΩ in parallel with the lower resistor.

The independent bias selector offers a 4.7 kΩ pull-up to port power, a 4.7 kΩ pull-up to 3.3 V, or a 4.7 kΩ pulldown. Every branch has its own resistor. The pulldown is a 0.75 W 2010 part because it can see a continuous 30 V signal. Pull-up diodes prevent signal voltage from feeding back through the bias circuit.

Industrial NPN normally uses the port pull-up; PNP uses the pulldown. Driven 3.3/5 V push-pull signals use logic thresholds and no bias jumper. A 3.3 V open-drain signal uses the 3.3 V pull-up and logic thresholds. Software inversion is separate from these electrical choices.

CAN and processing

An 8 MHz crystal and ×6 PLL supply the 48 MHz clock. The MCU samples four comparator outputs at 1 kHz, applies 1–100 ms selectable debounce, and reports digital state every 10 ms. Rising-edge counts and diagnostic/identity packets use slower bounded rates. Baseline periodic traffic is 131 frames/s, conservatively about 2.1% of a 1 Mbit/s bus per board.

The hardware filter accepts this device's Team Use address and reserves a separate receive FIFO for FRC Broadcast Disable. Broadcast Disable turns both sensor rails off and latches them; it does not stop diagnostic communication. Normal robot disabled-state heartbeat handling is not used to switch sensor power. No actuator commands are generated.

The UART service interface and SWD header remain available for bring-up and repair. Pin1 is a 3.3 V reference output, not a board-power input. The connector is a custom 2×5, 2.54 mm header and is not the standard Cortex 10-pin cable pinout.

Decisions changed during review

The battery switch changed from TPS1H100A to the stocked TPS1H200A, requiring a new pin map and exposed-pad footprint. Separate bias resistors replaced a topology vulnerable to multiple-jumper shorts. The filter increased to 1 nF. A third buck capacitor covers DC-bias loss. Q1 replaced a series diode to recover low-battery voltage margin. The final 101 kΩ feedback resistor centers the 5 V output with allowance for sensor-switch and PCB trace loss.

These are engineering choices supported by the checked files. Physical qualification is still required; see validation and bring-up.

3 Wiring and jumper settings

Source: docs/WIRING.md

The offline setup worksheet can produce a per-channel jumper table and a UART provisioning command for the common modes below. The two-page bench card shows the actual top-view connector orientation and six illustrated jumper examples.

Disconnect board power before moving jumpers or wiring terminals. Use the square pad, pin number and assembly drawing to identify pin 1; connector orientation on the page is not a pin-number definition.

Connector	Pin 1	Pin 2	Pin 3
J1, board supply	Positive 6–18 V	Ground	—
J2, J3, CAN trunk	CAN H	CAN L	—
J4, channel 1	Selected sensor POWER	Ground	SIGNAL
J5, channel 2	Selected sensor POWER	Ground	SIGNAL
J6, channel 3	Selected sensor POWER	Ground	SIGNAL
J7, channel 4	Selected sensor POWER	Ground	SIGNAL

J2 and J3 are electrically parallel. Continue the twisted CAN pair through these connectors, keeping untwisted sections and stubs short. Use the robot's common power ground; the CAN connectors themselves contain no ground pin. This board is not isolated. Fit JP2 only at a physical end of the CAN trunk. With the robot unpowered and all devices connected, two 120 Ω end terminations normally measure about 60 Ω between H and L.

Feed J1 from **one protected power-distribution branch**. A 5 A fuse branch with 18 AWG red/black copper input wiring is a conservative starting choice; select a fuse type allowed for the actual distribution device and season. The onboard 1.5 A fuse is additional board protection. Do not connect this board to battery terminals, the main unprotected distribution input, or the dedicated roboRIO/radio branch. Keep the PCB electrically isolated from the robot frame. Review the current FIRST robot electrical rules, including R610, R611, R615, R617, R620, R622 and R716; this project has no inspection approval.

The selected Wurth terminals specify 6 mm wire stripping and 0.4 N·m screw torque. Verify the current part drawing and use a suitable torque tool; do not leave exposed strands. Match sensor cable wire size and ferrules to the connector and sensor documentation. Brown/blue/black are common sensor power/ground/output colors, but **the sensor's own pinout controls**. Two-wire sensors need a separate compatibility assessment; these examples are for three-wire DC outputs.

Per-channel jumpers

Channel	Power selector	Threshold selector	Bias selector
1	JP101	JP102	JP103
2	JP201	JP202	JP203
3	JP301	JP302	JP303
4	JP401	JP402	JP403

Power selector: pins **1–2 = regulated 5 V**, **2–3 = protected battery**, open = no board power at that port. Install one shunt at most. Bridging all three pins joins two power rails and is prohibited.

Threshold selector: **open = industrial**, shunt fitted = **3.3/5 V logic**. Industrial mode targets LOW ≤4 V and HIGH ≥7 V. Logic mode targets LOW ≤0.8 V and HIGH ≥2.4 V. Voltages between each pair are not a defined

logic level; hysteresis intentionally retains the prior state over part of this interval.

Bias selector is a 2×3 header. Pin pairs are across each row: **1–2 = pull up to selected port power**, **3–4 = pull up to 3.3 V**, **5–6 = pull down to ground**. Install one horizontal shunt at most, or leave all open for a driven signal. Each branch contains 4.7 kΩ; pull-ups also contain a blocking diode. The assembly drawing marks pin 1.

Sensor/output	Power selection	Threshold	Bias	Typical software inversion
10–30 V three-wire NPN, active-low output	Battery, if sensor works over actual battery range	Industrial/open	Port pull-up, 1–2	Invert its channel bit
10–30 V three-wire PNP, active-high output	Battery, if compatible	Industrial/open	Pulldown, 5–6	Usually none
5 V push-pull	5 V, if power is needed	Logic/fitted	Open	Match desired meaning
3.3 V push-pull from another controller	Open; separate compatible supply	Logic/fitted	Open	Match desired meaning
3.3 V open-drain output	Compatible supply; 5 V only if sensor permits	Logic/fitted	3.3 V pull-up, 3–4	Usually invert
Dry contact to ground	Open	Logic/fitted	3.3 V pull-up, 3–4	Usually invert

The board does not provide a general 3.3 V sensor-power output. J8 pin 1 is a service reference, not a sensor supply. Selecting 5 V port power does not make a 3.3 V-only sensor safe to power.

For an externally powered 24 V sensor, connect its ground and output to the port's GND and SIGNAL, leave POWER disconnected, and leave the power selector open. A PNP output can use the onboard pulldown. An NPN output needs a suitable external pull-up to its sensor supply, or the onboard 3.3 V pull-up with logic thresholds if the sensor's off leakage and low-state voltage meet the logic limits. **Never feed external voltage into a POWER pin.** External supply arrangements must also comply with the robot rules.

The 300 mA limit is a design load budget for each entire rail, including all attached sensors and pull-up currents. It is not 300 mA per connector. A short or excessive startup current disables every sensor on that rail. Battery output is lower than board input by protection, switch and wiring losses; it is not regulated 12 V. Qualify low-battery behavior using the real sensor. A device needing at least 10 V may become unreliable well before this board resets.

Industrial examples assume low-state voltage ≤4 V and off leakage ≤0.5 mA; the NPN release calculation uses a 10 V port supply. The 3.3 V open-drain example budgets ≤50 µA off leakage and low voltage ≤0.8 V. Check the actual sensor data rather than its NPN/PNP label alone. Continuous positive SIGNAL voltage is limited to 30 V. The TVS does not make negative DC, AC mains, or higher continuous voltage acceptable.

Service connector J8

This is a **custom 2×5, 2.54 mm header**, not a standard Cortex debug cable pinout. Use individual labeled wires or a verified adapter.

Pin	Signal	Connection
1	3.3 V reference output	Programmer VTref input only; no external power
2	SWDIO	Programmer SWDIO
3	GND	Programmer ground
4	SWCLK	Programmer SWCLK
5	GND	UART ground, if convenient

Pin	Signal	Connection
6	UART TX from board	3.3 V USB-UART RX
7	BOOT0	Normally low; leave open
8	UART RX into board	3.3 V USB-UART TX
9	GND	Additional service ground
10	NRST	Programmer reset, if used

Leave the USB-UART power wire disconnected. Do not use a 5 V TTL output or an RS-232 voltage interface. JP1 briefly shorts NRST to ground for reset; do not leave a shunt installed. For ROM bootloader recovery, connect BOOT0 to the board's 3.3 V reference with a temporary wire, reset, use the documented STM32 UART bootloader, then remove the wire and reset. SWD programming is the primary supported procedure.

Software configuration and explicit fault recovery are described in firmware instructions. Do not configure a rail on until its wiring and jumpers have been inspected.

See representative sensor checks for NPN, PNP and 5 V break-beam examples and the extra supply/leakage checks each requires.

Representative sensor checks

No sensor was physically connected during this design session. These examples show how to apply the board's electrical limits to real products. They are separate from the PCB BOM and are not included in its cost. Confirm the exact purchased sensor's datasheet, cable and output behavior before use.

Example	Published electrical information	Board setup / remaining check
Omron E3Z-D61, NPN diffuse photoelectric	E3Z family: 12–24 V ±10%; 30 mA maximum consumption; open-collector output; residual voltage ≤1 V below 10 mA load; 1 ms response.	Battery power, industrial threshold, port-power pull-up. Confirm OFF leakage ≤0.5 mA for the actual revision. Sensor supply must remain ≥10.8 V at its cable end.
Omron E3Z-D81, PNP diffuse photoelectric	Same family supply/current envelope; PNP open-collector output.	Battery power, industrial threshold, 4.7 kΩ pulldown. Confirm OFF leakage ≤0.5 mA. At the stated supply minimum, the published residual-voltage allowance leaves enough HIGH-level margin.
Adafruit 2167, 3 mm IR break-beam pair	Published supply 3.3–5.5 V; receiver open collector; emitter draws 20 mA at 5 V; response <2 ms.	Regulated 5 V power, logic threshold, 3.3 V pull-up. Include both emitter and receiver in the rail-current budget. The product page does not establish a maximum total current or worst-case OFF leakage; verify them.

Sources: Omron E3Z specifications, Adafruit 2167 product details. The board adds its own 1 kHz sampling and debounce latency to the sensor response.

For a sensor on channel 1, the corresponding selector positions are:

Example mode	JP101 power	JP102 threshold	JP103 bias
E3Z-D61 NPN	2–3 battery	Open, industrial	1–2, pull to port power
E3Z-D81 PNP	2–3 battery	Open, industrial	5–6, pulldown
Adafruit 2167	1–2 regulated 5 V	Fitted, logic	3–4, pull to 3.3 V

Only one shunt belongs on each power selector and one horizontal pair on each bias selector. Select software inversion after observing the actual beam/object state; Light-ON/Dark-ON switches can change what an asserted transistor means. Do not infer logical game-piece presence from NPN/PNP alone.

The E3Z examples illustrate a battery-sag limitation: 12 V nominal does not mean the sensor remains within its rated supply during a match. The supplied intake-code example assumes a different generic sensor with a 10 V minimum and uses an 11 V diagnostic threshold. For a 10.8 V-minimum E3Z, use a more conservative starting policy such as **12,000 mV measured switched battery output**, then qualify actual cable drop, measurement error, hysteresis and sensor behavior. This is a robot-code policy example, not a calibrated undervoltage guarantee. A compatible 5 V sensor avoids this particular battery-following supply limitation.

For the break-beam pair, distribute the selected 5 V and ground to both halves using a suitable insulated splice or distribution connection; only the receiver's output goes to SIGNAL. Do not force multiple wire ends into a terminal unless that exact termination is qualified. Keep the cable mechanically secured and confirm operation under the venue lighting and intended target geometry.

DigiKey reference listings observed on 2026-09-07: E3Z-D61 / Z1108-ND, E3Z-D81 / Z5588-ND, Adafruit 2167 / 2167-ND. The D61 product page showed no immediate stock, the D81 showed 30, and the 2167 showed 934; older search listings disagreed. These are examples for engineering comparison, not an assertion that all are available to order now.

4 Purchasing and accessories

Source: docs/ACCESSORIES.md

These are reusable tools and mounting supplies, separate from the board's US$49.61 component estimate. Use equipment already owned when it meets the stated requirements. The accessory CSV records exact order codes and dated observations; nothing has been purchased.

Item	Selected part	Observed USD	Purpose
SWD programmer	STMicroelectronics ST-LINK/V2, 497-10484-ND	25.65	Flash and debug the STM32
Service jumper leads	Adafruit 1950, 1528-1961-ND	1.95	Twenty 150 mm female/female 2.54 mm jumper wires; use five for SWD
USB/UART cable	Adafruit 954, 1528-2128-ND	9.95	Configure board ID and startup settings using 3.3 V logic
Optional reusable hardware kit	Adafruit 4685, 1528-4685-ND	16.95	M3 nylon screws and standoffs; many boards per kit

The programming tools above total US$37.55 before shipping/tax. Including the optional hardware kit gives US$54.50. Stock observations on 2026-09-07 were 7,207 programmers, 10,997 jumper-wire packs, 5,225 UART cables and 78 hardware kits. Listings can change and the kit observation was an indexed distributor page. The manufacturer's own kit page showed no stock. Recheck the exact DigiKey order code before ordering.

ST-LINK/V2 to J8

J8 is a custom 2 × 5 header on **2.54 mm pitch**. It is not a standard Cortex 10-pin connector. Use individual female jumper wires, with power disconnected while attaching them. This mapping applies to the **non-isolated ST-LINK/V2 CN3 20-pin connector**, as defined by ST UM1075. Other probe connectors have different pin assignments.

ST-LINK/V2 CN3	Signal	Board J8
1, VAPP	Target voltage reference input	1, 3V3 reference output
7	SWDIO	2
9	SWCLK	4
15	NRST	10
4	Ground	3

Leave the probe's pin 19 supply output disconnected. Power the board through J1 from a current-limited supply; J8 pin 1 reports the board's voltage to the probe and must not be used to feed the board. Keep SWD wires short and start at 1 MHz adapter speed. The ST-LINK/V2-ISOL and STLINK-V3MINIE require their own verified connector mappings; a ribbon cable cannot be assumed to fit this header.

UART cable to J8

Adafruit 954 has 3.3 V TX/RX logic and a separate **5 V power lead**. Connect its ground to J8 pin 5 or 9, cable TX to J8 pin 8 (board RX), and cable RX to J8 pin 6 (board TX). Insulate and leave its power lead disconnected. Confirm lead identities against the supplied cable documentation; do not infer TX/RX from a generic cable's colors. Use 115200 baud, 8 data bits, no parity, one stop bit and no flow control.

Power the board before connecting an active UART transmitter so the service input is not driven while its 3.3 V supply is off. Disconnect UART and SWD leads before installing the board in the robot. The firmware guide contains the exact flash and provisioning commands.

Mounting and robot wiring

Use four M3 insulating standoffs at least 10 mm high, four M3 × 6 mm nylon screws on the PCB side, and suitable base-side fasteners. The optional 4685 kit includes these sizes. Check screw engagement in the actual standoff; avoid bottoming screws against one another. The 3.2 mm mounting holes are on an 88 × 88 mm square. Nylon mounting does not replace strain relief or an insulating cover.

Use stranded robot-rated wire sized for the actual protected power branch; 18 AWG is the recommended input wiring choice for this prototype. Use a twisted CAN H/L pair and the appropriate terminals at the robot's power-distribution device. Do not use the 28 AWG service jumper wires for robot power or sensor harnesses. Sensor cable gauge, flex rating and pinout depend on the actual sensor. Terminal strip length is 6 mm and nominal screw torque is 0.4 N·m for the selected Wurth connectors.

For qualification, provide a current-limited 0–30 V bench supply, multimeter, oscilloscope with short ground connection, adjustable load or suitable power resistors, and a 1 Mbit/s CAN interface or roboRIO. A 5 V 300 mA dummy load is approximately 16.7 Ω and dissipates 1.5 W; choose a resistor with adequate thermal margin and guard it against contact. A resistor load does not reproduce a sensor's startup capacitance or internal converter behavior.

The regional sourcing review records additional observations for critical PCB parts. Representative sensor examples are separate from both the board and accessory BOMs.

Regional sourcing review

The delivered PCB BOM contains exact DigiKey order codes for every purchased component. The design favors familiar leaded packages, common passive sizes and active products. Global availability cannot be guaranteed from a catalog listing; regional storefronts can use the same warehouse, show different cached inventory, and apply different shipping/import conditions.

The following additional checks support the availability preference for several critical parts. They supplement the dated US BOM rather than replacing it with mixed currencies or inconsistent timestamps.

Part	Additional source	Observed listing
STM32F042K6T6	DigiKey India	Indexed listing: 1,910 units, Active, ₹371.42 at quantity 1; 30-week manufacturer lead time.
STM32F042K6T6	Mouser UK	Indexed product listing: 3,350 units. Separate older/newer search snapshots differed; direct-page access did not complete.
STM32F042K6T6	ST online store	Manufacturer's indexed listing marked the T6 tray variant in stock.
LMR51420XDDCR	DigiKey Germany	Product page: 14,239 units, Active, €1.13 quantity 1 before VAT; 16-week manufacturer lead time.
TPS1H200AQDGNRQ1	DigiKey India	Product page: 25,389 units, Active, ₹170.08 quantity 1; 16-week manufacturer lead time.
TCAN3413DR	DigiKey UK	Product page: 4,877 units, Active, £1.38 quantity 1 before VAT; 16-week manufacturer lead time.

These are availability signals across regional catalogs and more than one seller, not separate stock pools, reserved supply or a worldwide-delivery promise. Long factory lead times still matter if distributor stock is exhausted. For a team build, check the exact quantities in the destination-country cart before paying. No cart, purchase or supplier contact was made in this session.

The MCU and several power/protection ICs remain single-manufacturer devices. Generic package familiarity does not make every similarly named part interchangeable. In particular, keep the TPS2553 **-1** latch-off suffix, LMR51420 **X** 500 kHz variant, LM2901 **B** grade, the exact TPS1H200A pinout, 100 V bias diodes, capacitor DC-bias allowance, and the 5.00 mm terminal pitch. BOM alternative fields identify review candidates only.

5 Hand assembly

Source: docs/ASSEMBLY.md

Use the interactive BOM, top assembly drawing and exact per-reference BOM. The board has 161 fitted components, ten bare test pads and four nonplated mounting holes. All fitted components are on the top side. Configuration shunts are installed only after inspection.

In the interactive BOM, filter by reference or manufacturer part number, then select a row to highlight its board position. Enable pin-1 highlighting when checking orientation. The 60 grouped assembly rows retain component-specific notes; the purchasing BOM combines these into 51 order lines. Widen the table with its divider or hide the Notes column when using a small screen. Sourced and placed checkboxes stay in that browser's local storage; they are personal assembly notes, not an inspection record shared with another computer.

Most passives are 0805 or larger, with visible leads on the MCU and most ICs. **U3 is the exception:** its HVSSOP exposed pad must be soldered to ground. A heatbed/reflow process with suitable paste and temperature measurement is needed for that joint. A soldering iron alone cannot reliably establish the underside joint. U3 has 0.65 mm lead pitch; U5 has 0.8 mm pitch.

Y1 uses Abracon's recommended 1.8 × 1.2 mm lands on 4.0 × 2.4 mm centers, extending beyond the nominal crystal body for solder access. Its electrical terminals are pads 1 and 3; pads 2 and 4 are grounded case contacts. Abracon permits the package chamfer at pin 1 or 4, so confirm the supplied part orientation against its drawing and markings rather than relying on a chamfer alone.

Recommended equipment: microscope or strong magnification, fine tweezers, temperature-controlled iron, flux, solder wick, a controllable heatbed/hotplate with thermocouple, optional hot-air assistance, multimeter, current-limited bench supply and oscilloscope. Use ESD handling and fume extraction appropriate for the chosen solder process. Buy paste from a traceable supplier and follow its storage and profile instructions.

Fabrication and stencil

Order a 96 × 96 mm two-layer board, 1.6 mm FR-4, 1 oz copper, with solder mask and lead-free HASL or ENIG. ENIG gives flatter small pads. The fabrication README lists drill and layer details. Request tenting of vias where the supplied mask does not open them. Do not add copper to the mounting-hole clearances or the C1 body keepout. Only C1's necessary footprint lands enter that body area; its ground land has a solid connection to the surrounding pour.

A 0.10 mm top stencil is the starting process choice; verify paste release with the assembler. U3's exposed-pad paste is split into four 0.65 × 0.8 mm windows, about 70% of the 1.57 × 1.89 mm exposed area. Four 0.30 mm thermal vias sit outside those paste windows. Do not replace the split apertures with one solid paste opening. Excess paste can float U3 or bridge its leads. The supplied bottom paste file is intentionally empty of component apertures.

Assembly sequence

1. Check board outline, solder mask, drill registration and pad finish. Confirm no visible shorts or broken tracks. Compare IC and connector footprints to the actual purchased parts before applying paste.

2. Apply paste to the top stencil and inspect every fine-pitch aperture. Place low-profile passives and ICs, then the crystal, inductor and aluminum capacitor. Keep through-hole headers and screw terminals out of the reflow cycle unless their exact manufacturer process explicitly allows it.

3. Check polarity and pin 1 before heating. U5 pin 1 is the package dot/chamfer reference, not the text direction. Q1's pin 4 is gate; pins 1–3 source face the protected rail and pins 5–8 drain face the fused input. TVS and bias-diode cathodes go to footprint pad 1. D3 cathode goes to VIN_PROT. C1 positive goes to VIN_PROT; the can's stripe marks negative. LED cathode is pad 1. Y1 pads 2 and 4 are grounded case pads.

4. Reflow using a measured board/component temperature profile compatible with the paste and the most restrictive component. **A heatbed setpoint is not the package temperature.** Do not invent a universal time/temperature recipe. Panasonic C1 requires its specified SMD reflow process; avoid prolonged heating.

Leave at least 2 mm above the capacitor vent in an enclosure. U3 is MSL 2, 260 °C classified, with one-year floor life under specified storage conditions. Follow actual bag labels and baking instructions if exposure history is uncertain.

5. Cool without moving the board. Inspect U1/U2/U3/U4 fine-pitch leads, all U5 pins, and diode orientation. Inspect for solder balls beneath the MCU and regulator. U3's underside joint cannot be fully verified optically; use process control plus electrical/thermal checks, or X-ray if available.

6. Iron-solder headers and screw terminals, keeping them square and fully seated. Use enough heat to wet ground-connected leads without cooking the connector body. Clean according to the flux manufacturer's instructions and dry completely.

7. Leave JP1, JP2 and all sensor selectors open. Perform the unpowered checks in BRING_UP.md before applying power. Add only the selected sensor shunts after rail validation.

The BOM's spare-parts import adds small passive/diode/LED spares. It does not buy extra ICs, connectors, crystal, inductor or fuse. Confirm R1 is **101 kΩ 0.1%**, R2 **13.7 kΩ 0.1%**, R4 **1 kΩ**, R14 **10 kΩ**, C14 **33 nF**, R5 **3.9 kΩ**, R10 **59 kΩ**, and the four pulldowns R109/R209/R309/R409 are **4.7 kΩ 2010, 0.75 W**. Do not substitute a small 0805 pulldown in that location.

Mount through H1–H4 using M3 insulating standoffs and hardware. Hole centers are (4,4), (92,4), (4,92), (92,92) mm from the board's top-left coordinate origin. Each has at least 3.5 mm radial copper clearance; the polygon vertices are at 3.52 mm radius. Keep metal, chips and wire strands away from the underside; provide cable strain relief and a ventilated insulating cover suitable for the robot. The STEP models are assembly envelopes, not enclosure-tolerance certification; terminal screws and wires need additional tool/bend clearance.

Record board serial number, PCB supplier/lot, component substitutions, paste/profile, inspection results and programming image hash. Any substitution affecting voltage, leakage, capacitance, pinout or package requires rechecking the relevant engineering calculation and physical test.

6 Firmware and provisioning

Source: firmware/README.md

This is a small bare-metal STM32F042K6 firmware, built with Arm GNU Toolchain 13.2.Rel1. ST CMSIS device files and Arm CMSIS headers are included with their licenses. The final two 1 KiB flash pages are reserved for redundant configuration records; the application linker permits 30 KiB of code and 6 KiB of RAM.

From the repository root:

```
source tools/env.sh
make -C firmware
bash scripts/test_firmware.sh
```

`firmware/build/sensor_can.elf`, `.hex` and `.bin` are the programming artifacts. The checked image occupies 7132 bytes of flash and 1832 bytes of reserved RAM, including a 1 KiB stack reservation. RAM figures do not establish measured worst-case stack depth. Host tests exercise the actual application and portable logic, but do not execute the STM32 peripheral registers.

Program and provision

Use an ST-Link compatible SWD probe with its target-power output disconnected. Power the board from a current-limited bench supply at J1. Wire the custom J8 header as listed in WIRING.md and the exact programmer mapping: VTref→1, SWDIO→2, GND→3, SWCLK→4 and NRST→10 if available. Do not plug a standard 10-pin Cortex cable into this header.

With OpenOCD 0.12 and an ST-Link probe, the intended command is:

```
openocd -f interface/stlink.cfg -f target/stm32f0x.cfg \
  -c 'adapter speed 1000' \
  -c 'program firmware/build/sensor_can.elf verify reset exit'
```

This command has not been exercised against a physical board. Confirm probe detection, target voltage, device identity and flash verification before relying on it. A normal application update should preserve the two configuration pages. A full-chip erase removes configuration and returns the firmware to its unconfigured behavior. Do not change readout protection or option bytes for ordinary bring-up.

Attach a **3.3 V logic** USB-UART adapter: adapter RX→J8.6, TX→J8.8, GND→J8.5 or 9. Leave its power lead disconnected. UART is 115200 baud, 8 data bits, no parity, one stop bit. Install `pyserial` for the host service tool.

```
python host/sensor_can.py info --port /dev/ttyUSB0
# Example: device 17, channel 1 active-low, 2 ms debounce, all sensor power OFF.
python host/sensor_can.py configure --port /dev/ttyUSB0 \
  --device 17 --invert 1 --debounce 2 --rails 0
```

The tool sends `SET`, waits for reset, then verifies persistent values through `INFO`. Device IDs are 0–63 and must be unique for this protocol. Inversion is a four-bit channel mask. Debounce is 1–100 ms. Rail masks are 0=off, 1=5 V, 2=battery, 3=both. A stored nonzero rail mask enables those rails automatically after an ordinary startup, so use rails=0 until wiring is verified. A watchdog reset overrides this with latched-off rails.

Once power and sensor checks pass, provision the desired rail mask explicitly, or keep the persistent setting off and request power through robot code. In a terminal, supported commands are `HELP`, `INFO`, `SET id invert debounce_ms rails`, and `CLEAR mask`, each followed by newline. `SET` disables rails, writes a CRC-protected record with its commit marker last, and resets. Two alternating pages retain the previous valid record if a write is interrupted. Flash endurance is finite: do not write persistent settings repeatedly from robot code.

After removing an overload and waiting at least 100 ms with the rail off:

```
python host/sensor_can.py clear --port /dev/ttyUSB0 --mask 2
```

This is a single recovery attempt. If a Broadcast Disable cleared the requested rail state, first request the desired rails through runtime configuration or a deliberate new persistent `SET`. Never continuously clear faults. The INFO response reports configuration, requested/enabled rails and fault state; a successful command is not proof of voltage at the sensor.

Behavior and diagnostics

With invalid or erased configuration, sensor power is off and the CAN transceiver stays in standby. Input processing and service UART still run. The 8 MHz crystal drives a 48 MHz PLL; CAN uses a 3×16 time-quanta arrangement for 1 Mbit/s and 87.5% sample point. A 1 kHz scheduler samples inputs, a configurable debounce filters state, and bounded queues prevent UART/CAN traffic from blocking indefinitely.

Hardware enable pulldowns and CAN standby pull-up establish reset defaults. Clock-security NMI, hard fault, low-voltage PVD and a foreground delay exceeding 20 ms force sensor power off. The independent watchdog then resets the device. Firmware enables the highest PVD level, whose falling threshold is nominally 2.78 V with a 2.66–2.90 V datasheet range. This protects logic operation; it does not keep a 10 V sensor working during battery sag. Sensor supply health remains an application responsibility.

The green status LED is a firmware heartbeat when CAN initialization succeeds; **it is not proof of an acknowledged or healthy CAN bus**. Red indicates unconfigured state, a clock/CAN initialization failure, ADC failure or latched sensor fault. Four green input LEDs indicate logical, debounced channel states. Review CAN diagnostics and stale-data handling even if the green LED flashes.

ADC calibration includes the STM32F042 erratum's delay before enabling conversion. VREFINT factory calibration compensates nominal VDD variation. ADC readings update every 1 ms; CAN diagnostic packets transmit every 100 ms. Both readings measure switched sensor outputs. Below a measured 4.5 V, an enabled rail latches off after the 5 ms startup blanking interval; ADC failure latches both rails. TPS1H200A can assert digital FAULT at a healthy 10–20 mA or lighter load, so firmware does not treat that ambiguous battery FAULT alone as an overload. Its hardware current-limit/latch remains active. Voltage readings are monitoring estimates and require physical calibration/accuracy checks.

The full packet layout, Broadcast Disable response and recovery semantics are in CAN_PROTOCOL.md. No wireless, USB, bootloader update protocol, CTRE protocol or actuator control is implemented.

Peripheral errata review

The STM32F042 errata were compared with the configured peripherals; the source-hashed review is in `reports/firmware-errata-review.json`. The ADC enable delay is present and CAN time-triggered mode is unused. The documented noisy-UART limitation remains a hardware concern; use short bench service wiring and verify configuration readback.

Stack review

The build emits GCC `.su` frame reports. `python scripts/audit_firmware_stack.py` combines these with the linked direct-call graph and inspected library prologues. The checked build estimates 212 bytes for the deepest foreground chain, plus a deliberately conservative 220 bytes counting all enabled/fault exception frames together. The 432-byte total fits the 1024-byte minimum linker reservation; 5336 bytes actually remain above static data. This static screen does not replace a hardware high-water-mark test or detect memory corruption. It rejects unreviewed indirect runtime calls, recursive call graphs and dynamic compiler stack frames. The emitted firmware binary is unchanged by stack-report generation.

Compiled ARM register-model tests

`python firmware/tests/test_arm_registers.py` executes the linked Cortex-M0 functions with Unicorn 2.1.4. Explicit synthetic register models provide readiness flags, ADC samples, CAN FIFO contents, UART buffering, flash erasure and GPIO set/reset effects. Thirteen check groups cover actual pin mux/modes, rail/fault/LED mappings, clock/PVD/watchdog setup, CAN filters/mailboxes, ADC scaling, UART overflow, and flash persistence including 34 modeled interruptions after halfword writes. This reaches the production `board.c` machine code that the host application tests replace with a fake board interface. It does not emulate STM32 peripheral timing, analog behavior, interrupt preemption, CAN arbitration or real flash brownout behavior. The physical bring-up tests remain required.

7 CAN protocol

Source: docs/CAN_PROTOCOL.md

The interface uses 1 Mbit/s Classical CAN with 29-bit extended data frames. It is an original Team Use protocol, with device type 10 (Miscellaneous) and manufacturer 8 (Team Use), following the WPILib addressing layout. It does not implement CTRE Phoenix or impersonate CTRE hardware. Coordinate IDs with any other team devices using this same type/manufacturer pair.

```
identifier = (10 << 24) | (8 << 16) | (api << 6) | device_id
```

`device_id` is 0–63, `api` is 0–1023. For example, device 17 status is `0x0A084011`. All team packets below have DLC 8. Multibyte integers are unsigned little-endian. Reserved bytes/bits must be zero. Invalid frame type, length, version, command magic, ranges or reserved bits are rejected. Receive filters reserve a separate FIFO for the standard FRC Broadcast Disable message.

API	Direction	Period	Bytes 0 through 7
0x100 status	Board → robot	10 ms	protocol=1, sequence, logical inputs, raw inputs, enabled rails, latched faults, health flags, reset flags
0x101 counts 1/2	Board → robot	100 ms	channel 1 uint32, channel 2 uint32
0x102 counts 3/4	Board → robot	100 ms	channel 3 uint32, channel 4 uint32
0x110 diagnostics	Board → robot	100 ms	switched battery output mV uint16, sensor 5 V mV uint16, TX queue drops uint16, TEC uint8, REC uint8
0x111 identity	Board → robot	1000 ms	protocol=1, firmware major=0, minor=1, patch=0, hardware=1, device ID, debounce ms, inversion mask
0x200 runtime configuration	Robot → board	On explicit change	0x53, 0x43, 1, inversion mask, debounce low byte, debounce high byte, requested rails, 0
0x201 clear fault	Robot → board	On explicit request	0x53, 0x43, 1, clear mask, 0, 0, 0, 0

Input bit 0 means channel 1, through bit 3 for channel 4. Raw inputs are comparator levels before inversion and debounce. Logical inputs are the stable, debounced states after XOR with the configured inversion mask. Rising-edge counters count changes to logical TRUE; they start at zero on boot and wrap modulo 2^{32}. Counter-pair frames are separate snapshots, not a four-channel atomic sample. Runtime configuration reinitializes debounce state while preserving counts. Input LEDs show logical state.

Rail bit 0 means regulated 5 V; bit 1 means battery. An enabled bit reports the MCU's commanded switch-enable state, **not measured power-good**. A hardware switch can have shut down before software updates the status. Latched faults require manual recovery. The battery diagnostic measures SENSOR_VBAT after the battery switch. The 5 V diagnostic measures the shared switched 5 V rail. Both are measured on the PCB; cable/connector loss and a disconnected sensor remain outside that measurement. Firmware samples both every 1 ms and latches an enabled rail off below a measured 4.5 V after its 5 ms startup blanking interval. ADC failure latches both rails off. The battery switch also asserts its digital FAULT for healthy loads below 10–20 mA, so battery fault handling uses output voltage while the switch retains autonomous current-limit/latch protection.

Health flags: bit 0 configured, bit 1 clock healthy, bit 2 ADC healthy. Reset flags: bit 0 power-on, bit 1 reset pin, bit 2 independent watchdog, bit 3 software reset; several bits can be set. Diagnostic ADC readings are approximate monitoring values, not calibrated instrumentation. Ignore voltage readings if ADC health is false. TX queue drops count local mailbox submission failures, saturating at 65535; they do not count every unacknowledged or corrupted transmission. TEC and REC are controller error counters.

Debounce is 1–100 ms with 1 kHz sampling. The default service example uses 2 ms. Analog filtering and sample phase add latency; seven typical-model SPICE cases observed roughly 2–3.1 ms transition latency with this setting. No pulse-width, quadrature, encoder speed or minimum-pulse guarantee is established. The status sequence increments modulo 256 and is a freshness aid, not a transaction identifier.

Runtime configuration accepts inversion 0–15 and rails 0–3. It is volatile and does not clear a latched rail. To recover a fault: remove the cause, leave the rail off for at least 100 ms, explicitly request the desired rail state, then explicitly clear its fault bit. Clear masks are 1–3. Do not retry automatically in robotPeriodic. UART SET is the only normal persistent configuration operation and resets the board after a verified flash write.

Timing and short pulses

Status is a periodic sample, so a brief debounced pulse can occur between 10 ms transmissions. Rising-edge counters retain such events until reset but arrive every 100 ms and have no event timestamps. Local debounce latency also excludes CAN and robot-loop delays. See the timing and pulse study before choosing a sensor geometry or event-handling strategy.

Broadcast Disable and startup

Extended data identifier **0x00000000**, with any DLC 0–8, is FRC Broadcast Disable. Firmware immediately commands both rails off, sets both latches and clears requested rails. Communication remains available. Recovery requires an explicit rail request and fault clear after the off-time requirement. Ordinary disabled-state heartbeat does not switch sensor power in this sensor-only design. This device has no actuator outputs.

With no valid flash configuration, CAN remains in standby and sensor rails remain off. Assign a unique ID through UART before connecting several unconfigured boards to a robot. A watchdog reset keeps both sensor rails latched off even if stored configuration requests them. Clock or low-voltage faults force safe shutdown. Automatic CAN bus-off recovery is enabled, but sensor data must still be treated as unavailable when stale.

Robot-side rules

Use a 100 ms status freshness limit; this is implemented in the supplied Java and C++ wrappers. Counter/diagnostic freshness is 350 ms and identity freshness is 1500 ms. Missing or invalid packets return no sample. Do not substitute a stale FALSE for a sensor failure. Verify configured, clock and ADC flags, the relevant rail enable/fault bits, and adequate sensor supply before acting. The supplied intake examples fail closed when their assumed 10 V sensor supply margin is absent.

There is no command acknowledgment packet, transaction ID, encryption or authentication. Read status and identity back after configuration. This is intended for a trusted, physically controlled robot CAN bus. Periodic traffic is 131 frames/s; a conservative 160 bits/frame estimate gives about 2.1% bus load per board. Team-controlled command rates should remain low.

8 Robot integration

Source: examples/README.md

The Java and C++ interfaces read this board's original Team Use CAN protocol. They compile against WPILib 2026.2.1 APIs. Host tests replace the transport boundary; an actual roboRIO integration test remains required.

For Java, copy `java/src/main/java/frc/robot/sensors/SensorCan.java` and `SensorCanCodec.java` into the matching package in a Java 17 robot project. Construct one `SensorCan` per physical board as a subsystem member, using the ID provisioned through UART. `status()`, `diagnostic()`, `counts12()`, `counts34()` and `identity()` return `Optional<Sample<...>>`. Missing, malformed or stale data returns empty. Close the object when disposing the subsystem.

For C++, copy the two headers from `cpp/include` into the robot project and construct `sensorcan::SensorCan` as a subsystem member. `ReadStatus()`, `ReadDiagnostic()`, `ReadCounts12()`, `ReadCounts34()` and `ReadIdentity()` return `std::optional<Sample<...>>`. The class uses the project's normal WPILib CAN library; no additional runtime library is required.

The intake examples in both languages assume channel 1 is logically TRUE when a game piece is present, with a battery-powered sensor requiring at least 10 V. They permit intake only when fresh status and diagnostics exist, configured/clock/ADC flags are healthy, battery power is enabled without a fault, switched battery output reports at least 11 V, and the piece is absent. The 11 V check is an example allowance for switch/wiring drop and diagnostic error; measure your actual sensor and adjust its policy deliberately. Neither example is a complete motor-control subsystem.

Reading data never enables sensor power or clears faults. `configureRuntime` / `ConfigureRuntime` sends one volatile configuration request; `clearFaults` / `ClearFaults` sends one explicit recovery attempt. Do not call either continuously from periodic code. Verify the resulting status/identity, and expose failures to the driver station. A constructor or successful packet write does not prove the board is present.

Status expires after 100 ms; counter/diagnostic packets after 350 ms; identity after 1500 ms. Counters are unsigned 32-bit totals and wrap. Java represents them as `long`; C++ uses `uint32_t`. Counter pairs arrive separately. Do not use these 1 kHz debounced inputs as high-speed encoders or rely on them as a safety interlock.

The Python service tool can also passively monitor a Linux SocketCAN interface already configured for 1 Mbit/s:

```
python host/sensor_can.py monitor --channel can0 --device 17 --seconds 30
```

Configure the CAN adapter using its own supported procedure. The monitor sends no commands and reports unavailable status after 100 ms without a valid frame. UART provisioning is described in firmware/README.md.

Run `bash scripts/test_drivers.sh` from the project root to compile and exercise Java, C++ and Python integrations using the installed toolchain. No LabVIEW VI is supplied; a LabVIEW implementation would need to implement the documented packets, freshness checks and fault policy, then be tested on the robot. This project does not claim complete ecosystem certification.

9 Bench qualification

Source: docs/BRING_UP.md

No physical board has been assembled or tested in this project session. Every item below is **pending** until someone records actual measurements. Start with the robot disconnected and use a current-limited bench supply. Save scope captures, board serial number, firmware hash, ambient temperature, load and instrument settings with the results.

Stage	Procedure	Initial acceptance target / action
Unpowered inspection	Check every IC pin 1, diode/capacitor polarity, U3 exposed-pad process, connector joints and jumper orientation against drawings.	No bridges, lifted pins, reversed parts or solder debris. All selectors and JP1/JP2 initially open.
Continuity	Verify J1 positive→F1→Q1 drain; Q1 source→VIN_PROT; grounds; CAN H/L connector continuity; each port's power/signal separation.	Match schematic. Rails must not show a persistent near-short to ground after capacitors charge from the meter. Resistance values alone do not prove correct assembly.
First input power	Set 12 V, initially 100 mA limit, no sensors/CAN/debug power injection. Apply J1 power and observe current and rail rise.	If current limit persists, power off and investigate; do not simply increase it. A correctly unconfigured board is expected to be well below the conservative 150 mA logic budget.
Internal rails	Measure TP2 VIN_PROT, TP3 V5_INTERNAL, TP4 3V3 relative to TP1 GND.	At room temperature, nominal about input minus small loss, 5.023 V, and 3.3 V. Initial DC targets 4.922–5.176 V internal 5 V, 3.201–3.399 V logic. Confirm startup has no damaging overshoot.
Default output state	Measure TP5 SENSOR_5V and TP6 SENSOR_VBAT with unconfigured firmware.	Enables low, rails off. Allow bleed resistors to discharge capacitors; a high-impedance meter may show leakage, so repeat with a modest test load.
SWD and firmware	Identify STM32F042K6, program supplied image, verify flash, read UART INFO.	Correct image and identity; configured=0 until explicit SET; no unexpected reset loop.
Clock/CAN timing	Observe a buffered timing output or CAN bit timing without loading the crystal directly; verify oscillator startup with a suitable low-capacitance method.	CAN 1 Mbit/s, stable startup over input/temperature. Characterize crystal drive ≤100 µW and frequency error; adjust R20/load capacitors only after analysis.
Rail enable, no load	Provision unique ID, first rails=0, then deliberately enable one rail at a time.	Correct rail responds, other remains off; UART/CAN state agrees with commanded enables.
Steady sensor loading	Electronic loads at 0, 50, 150, 300 mA on each rail, then both at 300 mA. Test J4–J7 individually and distributed loads.	Regulated port 5 V target 4.75–5.25 V steady at declared conditions; document battery drop. Logic remains healthy. Record connector and switch temperatures.
Battery sweep	Repeat loads at 6, 7, 9, 12, 14.4 and 18 V input; sweep down/up slowly through reset region.	No CAN lockup or spurious rail enable. Establish actual 5 V dropout and restart limits. Do not claim 6 V full-load regulation until measured.
Load steps	Scope internal/port 5 V and 3V3 for 0↔300 mA and mixed-rail steps; include real cable inductance.	Determine transient envelope and settling. A first-order estimate predicts about 0.10 V transient excursion; the ±5% steady target is not a guaranteed transient rating.
Current limit	Increase each electronic load beyond 300 mA while monitoring current, voltage, fault and enables.	Approximate 5 V limit 394–506 mA; battery limit 428–600 mA only where VS−OUT≥2.5 V. Confirm hardware latch-off and no automatic cycling.

Stage	Procedure	Initial acceptance target / action
Short and recovery	Use a controlled low-inductance short fixture on each rail, one at a time; capture current and time. Remove short, wait ≥100 ms, explicitly clear.	Battery nominal hardware delay about 11 ms; firmware may shut down earlier on measured output loss; 5 V overload deglitch specified 5–10 ms. Logic/CAN should remain operational. Verify a persistent short cannot be enabled indefinitely through software recovery.
Sensor startup	Enable all intended real sensors together at 6, 12 and 18 V where sensors permit. Include maximum cable/capacitance.	No nuisance latch; current and startup delay acceptable. Large capacitive loads may fail despite acceptable steady current.
Input thresholds	Use a current-limited adjustable signal source with common ground; ramp each channel up/down in both threshold modes at 6, 12, 18 V board input.	Logic: definite LOW at 0.8 V, HIGH at 2.4 V. Industrial: LOW at 4 V, HIGH at 7 V. Record actual hysteresis. Repeat at temperature endpoints.
NPN/PNP/leakage	Emulate NPN/PNP with off leakage up to 0.5 mA, ON residual voltage at intended bound, and actual sensor cable. Test 3.3 V open-drain with 50 µA off leakage.	Correct released/asserted states, all channel combinations, no cross-channel toggles. Check low-battery sensor behavior separately.
Filtering	Apply 49 µs glitches, then valid transitions at random sample phases; vary debounce 1, 2, 10, 100 ms.	Glitches rejected for characterized conditions; edge count and state follow intended debounce. Measure latency, do not infer high-speed encoder support.
CAN network	Connect a correctly terminated bench bus with another active node to provide ACK. Inspect H/L waveforms, errors and packet timing; then test roboRIO.	Correct ID/payload/rates, no dominant lockup, stable error counters, expected bus load. Confirm disconnect produces unavailable robot data within 100 ms.
Firmware stack watermark	In an instrumented test build, fill unused stack space with a pattern, exercise normal traffic, command floods and fault paths, then inspect the deepest overwrite.	Confirm adequate margin; the static 432-byte estimate is not a measured high-water mark. Preserve the test image hash.
Fault injections	Disconnect crystal only using a controlled test method, stall foreground in a test image, force watchdog reset, vary logic supply through PVD threshold.	Sensor enables fall; watchdog reset latches rails off; explicit recovery required. Do not short live MCU pins to simulate faults.
FRC Broadcast Disable	Send extended data ID 0 on a bench CAN network, including DLC 0 and 8.	Both rails off and latched; diagnostics continue; no automatic restart. Normal disabled robot state may retain sensor power.
Configuration interruption	On a dedicated test board, interrupt power at several points in a SET flash write, then reset.	Either old or new valid configuration loads; invalid records never enable an arbitrary rail/ID. Preserve captures and flash dumps.
Reverse polarity	With no external powered connections, apply −6 then −12 V through a conservative current limit for a short controlled test.	Protected rail remains unpowered; no heating or damage. Do not apply reverse voltage to sensor SIGNAL/POWER pins.
Unpowered inputs	Board off; apply supported positive sensor signals through a current-limited source; measure 3V3 and terminal leakage.	No unintended startup/back-powering, no damage; establish measured leakage.
Hot-plug	After basic operation, use a protected test fixture and actual harness at 6/12/18 V. Scope input current, VIN_PROT, Q1 VGS/VDS and rail overshoot.	Validate MOSFET/body-diode pulse stress, fuse survival and TVS energy. The passive model's ~113 A brief peak is not a qualified component stress result.
Thermal and environment	At maximum simultaneous load, soak at intended ambient, measure U1/U2/U3/U4/Q1/L1/C1 and connectors; then vibration/cable strain/EMI tests.	Junction and component temperatures remain below rated limits with margin. Establish a real operating-temperature rating; none has yet been qualified.

Use TP7–TP10 for comparator outputs INPUT1–INPUT4. Use short ground springs for switching-waveform probes. Avoid long probe ground leads that can invent ringing or short adjacent pads. Do not connect an

earth-referenced scope ground to a non-ground node. A differential probe is appropriate for Q1 gate-source and floating switch measurements.

Release gates before robot deployment

At minimum, complete polarity/continuity, rail regulation over actual battery range, load and short tests, all sensor modes, CAN/roboRIO freshness, reset/watchdog behavior, real-sensor startup and thermal checks. Record any reduced operating range discovered. Industrial EMC surge/EFT/ESD compliance and formal FRC approval are not established by this design or its simulations. Keep the first prototype accessible for measurements and repair.

Suggested measurement log columns: date, board serial, firmware SHA256, test name, input voltage, rail loads, sensor/model/cable, ambient, measured min/max, scope file, pass/fail, corrective action. A CSV template is provided in `reports/bench_measurements_template.csv`.

The fillable test plan lists every stage above with status PENDING and blank result/evidence fields. Copy it for each physical board and retain failed results alongside any corrective action. The separate measurement template supports multiple readings and operating conditions per stage.

10 Electrical analysis

Source: docs/ELECTRICAL_REVIEW.md

The authoritative circuit is `hardware/design/circuit.py`, exported into the native KiCad schematic. The final PCB is checked against its exported netlist and this model. This review describes the actual 171-symbol / 161-purchased-component revision, including Q1 reverse-polarity protection and the final 101 kΩ feedback resistor.

Input levels and component stress

Each channel divides SIGNAL through 220 kΩ total into 20 kΩ to ground, with 910 kΩ comparator feedback. The LM2901B is powered from VIN_PROT and its open-collector output is pulled to 3.3 V. Logic thresholds use the parallel 7.15 kΩ reference resistor; industrial thresholds leave that branch open. Two series 110 kΩ resistors distribute signal-voltage stress.

The corner script enumerates 8192 combinations per mode, using ±1.75% total ordinary-resistor allowance, ±3% 3.3 V supply, ±5.5 mV comparator offset, ±50 nA input bias, output LOW up to 0.55 V and 100 nA high-state leakage. The resistor allowance includes 1% initial tolerance, 100 ppm/°C across a 65 °C excursion, and an additional 0.1% allowance. It is an engineering budget, not a qualified board temperature range.

Mode	Calculated rising threshold	Calculated falling threshold	Declared prototype logic targets
Logic	1.571–2.240 V	0.938–1.425 V	LOW ≤0.8 V, HIGH ≥2.4 V
Industrial	5.352–6.779 V	4.718–5.960 V	LOW ≤4 V, HIGH ≥7 V

The 4.7 kΩ 2010 pulldown dissipates at most about 0.195 W at 30 V with the resistor allowance, against a 0.75 W nominal rating before ambient derating. A PNP output leaking 0.5 mA produces about 2.39 V at worst, below the industrial LOW target. An NPN output with 0.5 mA leakage and a 10 V port pull-up releases to at least about 6.87 V in the model. A 3.3 V open-drain output with 50 µA off leakage releases to about 2.315 V under the selected diode/supply allowance. The pull-up calculation uses a 0.60 V engineering diode-drop allowance; the released open-drain value remains above the calculated 2.240 V maximum logic rising threshold. These bounds depend on actual sensor residual voltage, leakage and diode temperature behavior.

The 30 V standoff TVS is transient protection, not a continuous regulator. Its dynamic clamp can reach 48.4 V under its specified pulse condition; the series divider keeps the comparator input much lower. The unidirectional TVS conducts strongly for negative signal voltage, so sustained negative/AC input is not supported. Bias diodes are 100 V BAT46W parts; a common 30 V BAT54 is not an equivalent substitution.

Seven transient cases use TI's LM2901 macro-model under ngspice, with a narrow PSpice area-parameter syntax adaptation retained alongside the original. They cover 3.3/5 V logic, 12/30 V industrial, NPN/PNP leakage and 3.3 V open-drain. A 49 µs glitch is rejected and a valid transition counted once at all 100 sample phases per case using the production C debounce code. This is 700 modeled phase runs. Typical-model transition latency at 2 ms debounce is approximately 2–3.1 ms. Models do not establish worst-case comparator delay, cable EMC immunity or a guaranteed pulse-width specification.

Power regulation and distribution

The selected LMR51420XDDCR is the 500 kHz PFM version. Its reference/regulation budget and 101 kΩ / 13.7 kΩ feedback pair yield 5.023 V nominal, with a calculated DC interval 4.922–5.176 V including ±0.3% total feedback-resistor allowance. Do not substitute the Y variant (different frequency) without reviewing L/C and loop behavior.

Three Samsung CL31A226KAHNNNE 22 µF/25 V output capacitors provide 66 µF nameplate capacitance. The manufacturer's typical DC-bias curve loses about 39.5% at 5.176 V. Applying tolerance, temperature, aging and extra modeling allowances gives about 26.10 µF effective output capacitance, above the 25 µF design target. This is a typical-curve engineering estimate, not a manufacturer-guaranteed minimum. An 8 µH minimum inductance and 450 kHz minimum frequency produce about 1.006 A peak-to-peak ripple at 18 V

input. L1 has 6 A stated saturation current, above the selected 4.2 A regulator stress criterion.

The first-order 300 mA load-step estimate is about 0.102 V. Therefore the 4.75–5.25 V sensor target is a **steady-state target**; full transient compliance remains a bench test. The regulator is PFM at light load. Check ripple, overshoot, dropout and load-step settling with actual capacitors and PCB parasitics.

The actual routed positive power paths were analyzed using copper geometry at 85 °C, 35 µm copper and a 20 µm via-plating assumption. The graph uses one shortest resistive path and ignores parallel copper benefits; pads are lumped and ground-plane/connector resistance is an added allowance. The longest 5 V rail route is about 103 mm and the longest battery route about 162 mm. With 300 mA sensor current plus a conservative 10 mA bias/bleed allowance concentrated at one connector, U4's 135 mΩ hot maximum, 460 mA upstream budget and 60 mΩ additional contact/shunt/return allowance, predicted minimum regulated connector voltage is **4.806 V**. This calculation is not a field-solved thermal/current-density result.

At 6 V input with both rails supplying 300 mA, 150 mA conservative 5 V logic budget and 80% buck efficiency, estimated input current is 0.830 A including additional 20 mA battery and 10 mA regulated-rail bias/bleed allowances. A 0.196 Ω total entry resistance allowance gives 5.837 V at VIN_PROT. Q1's estimated dissipation is 27.5 mW. The 40 mΩ hot Q1 allowance is not a guaranteed hot resistance at every gate voltage. AP2112 dissipation at the full conservative logic budget is about 0.281 W and requires thermal verification on this PCB.

Fault protection and startup

The 5 V switch's 59 kΩ current-limit resistor gives approximately 394–506 mA over the selected equation/tolerance budget. Its overload fault deglitch is 5–10 ms and it latches off. The battery switch's 3.9 kΩ setting gives approximately 428–600 mA; specified accuracy requires VS−OUT≥2.5 V. The 33 nF delay capacitor gives about 11.03 ms nominal hardware latch delay, using typical timing-current/reference values. Actual timing must be measured. Both rails are intended for 300 mA total continuous sensor load, not for actuators.

At 18 V, 40 µF external capacitance plus a 1.1 µF onboard allowance and 320 mA steady load including bias/bleed takes about 6.82 ms to charge at the minimum calculated current limit. The longer 33 nF timing choice improves margin for sensor startup. Timing parameters are typical, so this is not a guaranteed capacitance rating. Sensor startup current can exceed steady current, and larger loads may latch off. There is no automatic retry. Hardware pulldowns keep rails off during reset; software provides manual recovery, watchdog-reset latching and Broadcast Disable handling.

Q1 AO4485 is a 40 V P-channel MOSFET with its drain toward the fused input. D3 clamps its gate-source voltage near 12 V. The topology protects reversed input polarity, but an enhanced MOSFET conducts both directions; it does not block an externally powered output from feeding upstream. External POWER-pin injection is prohibited.

F1 is a 1.5 A fast Littelfuse 451 with nominal 0.853 A²s melting integral and 300 A interrupt rating at 32 V DC. It supplements the protected robot branch; it is not a substitute for upstream distribution protection. SMBJ18A has 18 V standoff and 29.2 V specified clamp at its test current. Continuous board supply is limited to the 6–18 V target; the TVS does not authorize sustained higher voltage.

A 54-case passive hot-plug sweep varies 6/12/18 V, source resistance, harness inductance and bulk-capacitor ESR. It predicts up to about 25.32 V at VIN_PROT, 9.90 mJ TVS energy and 0.0797 A²s input integral in the selected scenarios. A brief input-current peak reaches about 112.75 A. The model treats Q1 as already on and does **not** model gate charging, body-diode safe operating area, fuse temperature, switch failures or standardized surge waveforms. It is a sensitivity study and a reason to measure hot-plug stress, not proof that protection is qualified.

Clock, CAN and assembly constraints

The final review corrected the battery switch's open-load interaction: its FAULT output can assert below 10–20 mA even when output voltage is healthy. R23 now senses SENSOR_VBAT, and firmware samples both switched outputs every millisecond. A measured output below 4.5 V latches that enabled rail off after 5 ms startup blanking; ADC failure latches both. The battery switch's autonomous current/thermal protection remains active. Tests specifically cover a healthy asserted open-load FAULT, true output loss with and without FAULT, startup ramp/timeout and runtime ADC failure.

R4 is 1 kΩ and R14 is 10 kΩ. Using BAT46W's 15 µA maximum leakage at 50 V/60 °C as an allowance for four positive signals up to 30 V, plus a conservative feedback-current estimate, the resistor model predicts at most about 0.075 V on unpowered 3V3 and 0.560 V on the battery output. These are leakage-path estimates, not guaranteed powered-off behavior or a 125 °C leakage specification. The qualification target is 0–60 °C ambient; no operating-temperature range has yet been physically established. Characterize the comparator, contamination, cables and power sequencing on real hardware.

The 8 MHz crystal has 18 pF specified load. Two 30 pF capacitors assume approximately 3 pF parasitic load. The calculated transconductance startup margin is about 7.9 using the datasheet ESR/C0 bounds and MCU minimum drive parameter. Verify startup and ≤100 µW crystal drive physically. R20 permits later series-drive tuning.

TCAN3413 uses 3.3 V logic and supply, with standby and TXD pull-ups. The separate ESD2CAN24 protects H/L. Optional 120 Ω termination is open by default. Ground is common with the robot; the bus has no galvanic isolation. A 1 Mbit/s timing configuration of BRP=3, BS1=13, BS2=2 at 48 MHz yields an 87.5% sample point. Board routing checks do not establish bus EMC compliance.

U3's exposed pad is grounded through copper and four thermal vias. Its footprint and stencil windows were built from the selected DGN package drawing. All part pin mappings are explicit in the source model; KiCad ERC, PCB DRC and model/schematic/board parity checks pass. Those checks detect modeled connectivity/clearance errors, not incorrect datasheet interpretation, solder defects or environmental failure. See VALIDATION.md for the evidence boundaries and BRING_UP.md for remaining physical tests.

Buck layout and simulation scope

The colored power-layout drawing shows actual exported copper on both layers, viewed from the top. C3 connects directly to the adjacent U1 VIN/GND pins. The final VOUT feedback pickup is at C13's positive terminal; its bottom-layer route stays outside the inductor body. The high-impedance FB node remains short on the top layer beside R1/R2. This follows the regulator's placement guidance but does not establish EMI performance; the prototype still needs ripple, load-step and emissions checks.

The actual X-variant TI transient model, SLUM848, was downloaded and inspected. It is encrypted for PSpice 17.4 and cannot run in the installed ngspice engine. No closed-loop buck transient simulation is claimed. The archive and its features/limitations document remain in `sources/models/lmr51420x_ti`. Its header also notes a shortened simulated hiccup interval and no temperature-dependent behavior. The calculated capacitance/ripple/load-step allowances still require the physical tests in the bring-up plan.

Ground return and manufacturing geometry

A separate DC resistor-sheet model reconstructs the GND copper from the exported Gerbers, then joins the two layers through the 85 ground-connected plated barrels. It assumes 35 µm copper at 85 °C, 20 µm barrel plating and equipotential top-side contact pads. The solver passes an analytic rectangular-sheet calibration and reciprocity/residual checks. Meshes of 0.15 mm and 0.10 mm agree within 4.4% for the examined routes; the finer mesh solves about 1.26 million unknowns.

Estimated sensor-ground resistance to the buck output-capacitor ground is 3.46–3.57 mΩ, corresponding to about 1.11 mV at 310 mA on the worst examined port. This supports, but does not replace, the existing 60 mΩ contact/shunt/return allowance. Mesh-center edge approximations, filled pad interiors, contact placement and nominal copper/plating dimensions limit accuracy. It models no switching inductance, ground bounce, thermal rise, EMC or manufactured defects. See `reports/electrical/ground_return_model.json` and its two plots for the assumptions and exact Gerber hashes.

Independent exported-geometry checks also find minimum different-net clearance of 0.200 mm, copper-to-board-edge clearance of 0.5005 mm, mounting-center clearance of 3.512 mm, and approximately 0.150 mm minimum plated-hole annular copper. The minimum hole-to-hole web is 0.558 mm. Arc polygons have a 1 µm approximation tolerance. Top mask openings have at least 0.200 mm between separate openings; bottom component paste is empty. These checks screen nominal files and do not guarantee a fabricator's registration, drill wander, plating or stencil process. See `reports/manufacturing/gerber_geometry.json`.

The final layout clears a 3.60 mm nominal-radius region under C1's can on both copper layers, with only its footprint lands permitted. Its VIN via and connecting traces exit outside that body projection. This follows Panasonic's guidance to avoid circuit patterns beneath the aluminum capacitor. C1's ground land connects solidly to the surrounding ground pour; the ceramic input capacitor remains close to U1 for the high-frequency switching loop.

11 Validation evidence

Source: docs/VALIDATION.md

The editable design, exports and software have been checked computationally. **No physical PCB, sensor, CAN adapter or roboRIO was available for bench validation.** Passing software/design checks does not establish electrical ratings, EMC compliance, assembly yield or competition approval.

Check	Result	Evidence and scope
KiCad schematic ERC	0 violations	`reports/erc/final.json`; electrical-rule checking of the modeled schematic
KiCad PCB DRC	0 violations, 0 unconnected items	`reports/drc/final.json`; copper clearance, connectivity and configured manufacturing rules
Circuit/schematic/PCB parity	PASS	`reports/design_parity.json`; 171 modeled parts, 99 nets, 439 schematic pin nodes; 443 board pads including four additional U3 thermal pads
Final device pin review	No assignment mismatch found	`reports/electrical/pin_review.json`; manual manufacturer-source comparison of 97 pins across 11 components; crystal lands changed to Abracon's recommendation
Independent Gerber parsing	PASS	`reports/manufacturing/verification.json`; Gerbonara verifies 443 top and 80 bottom X2 pad/net/position records, 268 plated drills, 4 nonplated drills, 96 mm outline, 161 placements and purchased references
Independent Gerber copper connectivity	PASS	`reports/manufacturing/gerber_connectivity.json`; separate polygon reconstruction and plated-barrel joins find no shorts, opens or orphan copper islands across 99 named nets
Independent Gerber geometry	PASS	`reports/manufacturing/gerber_geometry.json`; clearance, board edge, mounting clearance, annular copper and drill web checks; mask/paste metrics
DC ground-return estimate	Numerical checks pass	`reports/electrical/ground_return_model.json`; analytic sheet calibration, reciprocity, solver residuals and two mesh sizes; DC approximation only
Local mechanical models	PASS	161 fitted-component model paths resolve inside the project; simplified original envelopes are identified separately
DigiKey purchasing completeness	PASS	51 purchase lines, 161 fitted components plus 13 shunts, no zero-stock selected line in dated observations; not a live quote
DC electrical corners	PASS	`reports/electrical/calculations.json`; 8192 input corners per threshold mode, rail/current/stress/clock calculations
Capacitor and routed-power allowances	PASS under stated assumptions	`reports/electrical/capacitor_derating.json`, `power_routes.json`; typical DC-bias curve and geometric resistance estimates
Input transient simulation	7 cases × 100 sample phases pass	`reports/spice/input_results.json`; TI macro-model plus actual production debounce code
Pulse capture characterization	14 waveforms × 84 widths × 4 settings × 100 phases	`reports/spice/pulse_response.json`; expected short-pulse misses are characterized, not a hardware minimum-width guarantee

Check	Result	Evidence and scope
Passive hot-plug sensitivity	54 cases completed	`reports/spice/inrush_results.json`; useful stress estimates, not MOSFET/TVS/fuse qualification
ARM firmware build	PASS, warnings treated as errors	`reports/firmware-build.log`; 7132 bytes flash, 1832 bytes reserved RAM in checked build
Static linked-firmware stack screen	PASS	`reports/firmware-stack.json`; 432-byte conservative direct-call/exception estimate within 1024-byte minimum reservation; no runtime watermark measurement
Compiled ARM register-model execution	13 groups pass	`reports/firmware-arm-registers.json`; actual Cortex-M0 binary functions, GPIO/CAN/ADC/UART/watchdog/flash models and 34 halfword interruption cases; no peripheral timing or silicon emulation claim
Portable firmware logic	2,190,989 assertions pass	`reports/firmware-tests.log`; address/undefined-behavior sanitizers enabled
Actual application boundary tests	18 scenarios pass	Same log; normal/unconfigured/clock/CAN/ADC/watchdog/rail-fault/bad-command/UART/stall/busy plus light-load, output-loss and startup cases with simulated board boundary
Java integration	232,928 assertions pass	`reports/driver-tests.log`; Java 17, real WPILib 2026.2.1 jars, simulated CAN transport
C++ integration	75,828 assertions pass	Same log; C++17, actual WPILib 2026.2.1 headers, simulated CAN transport, sanitizers
Python service/protocol	5 tests pass	Same log; protocol vectors, all IDs, random counters, actual UART CLI over a simulated PTY board
Offline setup worksheet	PASS in browser	`reports/worksheet_review.json`; mixed-mode jumper/command checks, budget/range rejection, desktop/mobile and print review; no hardware connection
Interactive assembly BOM	PASS in browser	`reports/assembly_browser_review.json`; all 161 reference/MPN pairs match the purchasing CSV, component/net lookup, pin-1 highlighting and checkbox persistence
Physical electrical, thermal and robot tests	PENDING	Follow `docs/BRING_UP.md`; no measured hardware claims

The interactive-BOM exporter warns that H1–H4 are absent from the schematic. These are intentional mechanical mounting holes, excluded from electrical netlist/BOM. The separate parity and manufacturing checks account for them explicitly. Gerbonara reports benign KiCad-format/parser warnings concerning header command placement and unused aperture-macro parameters; these are retained in its report rather than silently discarded. No missing copper pads or drill mismatches remain.

The 3D render and Gerber previews are visual review aids. Standard package models and six original simplified package envelopes are included locally. Nominal models do not replace supplier tolerances, wire-bend clearance, tool access or an enclosure fit test. Native KiCad files remain the physical design authority.

Reproduce the checks

With the documented toolchain installed, run `bash scripts/validate_release.sh`. This regenerates electrical/BOM data, checks the native design, builds/tests firmware and drivers, exports manufacturing files, and independently compares those exports. SPICE runs are included in that script and save their assumptions/results. Read `TOOLS.md` for installation/provenance and `hardware/design/README.md` before running historical layout scripts.

Do not interpret a check that was not run as passed. The release manifest records actual output hashes so later changes can be distinguished from this reviewed revision. A later footprint, value, firmware or package substitution invalidates the affected result until its relevant checks are rerun.

12 Sources and notices

Source: docs/SOURCES_AND_LICENSES.md

Sources were consulted on 2026-09-07. Exact purchase SKUs, product-page links, stock observations and quantity-one prices are stored in `bom/catalog_selected.json` and `BOM_grouped.csv`. They are dated distributor observations, sometimes from indexed listings, not reserved inventory or proof of worldwide delivery. All fitted purchase lines use DigiKey order codes. Alternative MPNs are candidates for engineering review, not approved substitutions.

Primary technical references

Subject	Primary source	Used for
Functional reference	CTRE CANdi, user guide	Sensor-to-CAN role and regulated/battery sensor outputs; no inherited ratings/protocol
Robot electrical rules	FIRST 2026 manual	Protected power branch, wiring, isolation and CAN constraints; recheck season rules
CAN addressing	WPILib CAN device addressing	Team Use manufacturer, device type, Broadcast Disable
Robot CAN API	WPILib 2026.2.1 source, Java CAN API	Actual Java/C++ transport interfaces
MCU	STM32F042x4/x6 datasheet, ES0243 errata	LQFP32 pin map, voltage/clock/ADC limits, calibration delay
CAN PHY	TCAN3413 datasheet	SOIC8 pin map, supply, standby and bus protection
Buck regulator	LMR51420 datasheet	X=500 kHz PFM variant, L/C, reference, layout and current limits
Logic regulator	AP2112 datasheet	SOT25 pin map and output-capacitor requirements
Battery switch	TPS1H200A-Q1 Rev. E	DGN pin map/land pattern, limit, latch delay, diagnostics
5 V switch	TPS2553 datasheet	DBVR-1 latch variant, current-limit equations, fault timing
Comparator	LM2901B datasheet, TI simulation model	B-grade electrical limits, quad pin map, transient macro-model
Reverse protection	AO4485	MOSFET pinout, gate/drain ratings, resistance and pulse limits
Fuse	Littelfuse 451/453 datasheet	1.5 A package, interrupt rating, resistance and nominal melting integral
Bias diode	BAT46W	100 V reverse rating, forward drop and package
CAN protection	ESD2CAN24-Q1	SOT23 H/L/GND mapping and bus protection
Inductor	SRP7028A	10 µH current ratings and mechanical outline
Crystal	ABM3B	8 MHz, 18 pF load, ESR/C0, drive level and outline
Connectors	691137710003, 691137710002	Exact 5 mm pitch, pin geometry, body, strip length and torque
Capacitor derating	Samsung CL31A226KAHNNNE manufacturer characteristics saved in `sources/datasheets/samsung_CL31A226_characteristics.json`	Typical DC-bias curve used by calculation; not guaranteed minimum capacitance
Aluminum capacitor	Panasonic FK series	EEE-FK1V470P dimensions, ripple/impedance and handling cautions
ESP32 comparison	Espressif ESP32-C3 TWAI	CAN controller still requires an external transceiver

Local datasheets and extracted text are research references. Files for rejected parts remain in `sources/datasheets` to preserve research history; only the current BOM and circuit determine what to purchase. `sources/source_manifest.json` records local-file hashes and provenance where available. Manufacturer documents and models retain their owners' copyright and terms; they are not relicensed as original project material.

Libraries and software

KiCad standard footprints and package models came from the verified KiCad 10.0.6 distribution. Their original names are retained in each part's `source_footprint` field. Project-local copies make the design portable. The KiCad library license is CC BY-SA 4.0 with its design-output exception; the notice and legal text are retained in `licenses`. Original custom footprints cover the selected Wurth terminals and TI exposed-pad package. The Abracon crystal footprint adapts the KiCad body/model to the manufacturer's recommended land pattern, with its derivation recorded in the footprint provenance file. Six original simplified STEP models are listed with drawing sources in `hardware/library/3d/original_models.json`. They are not supplier-certified models.

Firmware includes ST CMSIS device headers/startup and Arm CMSIS headers with their original license files. WPILib headers/jars are used for integration compilation and retain the WPILib notice. InteractiveHtmlBom generates the standalone assembly page; its license is included. TI's original LM2901 model and copyright header are preserved, and the ngspice-compatible adaptation changes only the diode area-parameter syntax needed by the simulator. See `licenses/` and the respective third-party directories for the complete notices.

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